How AI Sees KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

1. Color by number. Color every square with a **1**. Leave every **0** white.

0	0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0	1	1	0	0
0	1	1	1	1	0	0	1	1	1	1	0
1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1	1	0
0	0	1	1	1	1	1	1	1	1	0	0
0	0	0	1	1	1	1	1	1	0	0	0
0	0	0	0	1	1	1	1	0	0	0	0
0	0	0	0	0	1	1	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0

What is the picture? a heart

2. Pixels are numbers.

In a black-and-white picture, 0 is black and 255 is white.

In a color picture, each pixel is 3 numbers: red, green, and blue.

3. Find the edges on screen.

With the **up-and-down** filter, the bright squares are on the walls of the house.

I clicked a bright square. The answer was Answers vary. I clicked a dark square. The answer was 0.

With the **side-to-side** filter, the bright squares are on the top and bottom edges.

A big answer means the picture changes fast here: an edge.

4. Talk about it. What could a self-driving car miss?

A self-driving car could make a mistake when Answers vary. It sees something new.

Computer vision is helpful when Answers vary. and dangerous when _____

Go Further (Day 2). Find the edge by hand. Here is a tiny picture (0 = dark, 9 = bright) and a filter.

0	0	0	9	9	9
0	0	0	9	9	9
0	0	0	9	9	9

Filter:

−1	0	1
−1	0	1
−1	0	1

Put the filter over 3 columns of the picture. Multiply each pair of numbers, then add all 9.

Filter over columns	Math	Answer
1, 2, 3	each row: $-0+0+0=0$	0
2, 3, 4	each row: $-0+0+9=9$, so $9+9+9$	27
3, 4, 5	each row: $-0+0+9=9$	27
4, 5, 6	each row: $-9+0+9=0$	0

Where are the big answers? Where the picture changes from 0 to 9: the edge.